E Finetuning Setup

The total loss is initially comprised of three terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \alpha\mathcal{L}_{\text{reg}} + \beta\mathcal{L}_{\text{cls}}$$

We finetune on the training set from §B.1 using LoRA (Hu et al., 2022) with settings `lora_r=16`, `lora_alpha=32`, `lora_dropout=0.1`, `target_modules=[q_proj, v_proj]`. We use AdamW (Loshchilov and Hutter, 2019) as our optimizer, finetuning for 3 epochs with batch size 16.

We first initialize the probe parameters by fitting them in our usual way (§D) on the same training data, before proceeding with finetuning of all parameters as above.

Using the validation set from §B.1, we perform grid-search on probe and finetuning hyperparameters: $\alpha, \beta \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 50, 100\}$, learning rate $\in$ {`2e-6`, `5e-6`, `1e-5`, `2e-5`, `5e-5`, `1e-4`, `2e-4`}, and probing layer depth $\in$ {`10%`, `20%`, `30%`, `40%`, `50%`, `60%`, `70%`, `80%`, `90%`, `100%`}. We select the hyperparameters that achieve the highest verbalization accuracy on validation data.

These optimal hyperparameters are $\alpha = 0$, $\beta = 0.02$, learning rate = 5e-5, probing layer depth = 90%. Thus the regression probing loss term is omitted in §2 since including it is suboptimal. In fact, we found that $\alpha > 0$ was suboptimal even when $\beta = 0$, indicating that attempting to improve regression loss did not help with the verbalized comparison. Presumably that was because regression on the `cross-notation` dataset was already extremely accurate. On a dataset like `arXiv`, however, there would be more room to improve regression, and this might benefit the downstream task of verbalized comparison.

F Evaluation Metrics

Mean squared error (MSE) is the average squared difference between probe predictions and actual values in log space.

Relative error (RE) (reported in our paper abstract) converts the median absolute error in log space to a relative error as shown below.

Approximate accuracy (AAcc) evaluates whether the predicted number is approximately the same as the original number in normal (non-log) space, namely with an error margin of $< 1\%$. Higher AAcc indicates that the number encoding is more likely to be precise.

$$\text{MSE}(\boldsymbol{y}, \boldsymbol{x}) = \text{mean}((\boldsymbol{y} - \log_2 \boldsymbol{x})^2) \tag{1}$$

$$\text{RE}(\boldsymbol{y}, \boldsymbol{x}) = 2^{\text{median}(|\boldsymbol{y} - \log_2 \boldsymbol{x}|)} - 1 = \text{median}\left(\max\left(\frac{2^{\boldsymbol{y}}}{\boldsymbol{x}} - 1, \frac{\boldsymbol{x}}{2^{\boldsymbol{y}}} - 1\right)\right) \tag{2}$$

$$\text{AAcc}(\boldsymbol{y}, \boldsymbol{x}) = \frac{\left|\, |2^{\boldsymbol{y}} - \boldsymbol{x}| < 0.01\boldsymbol{x} \,\right|}{|\boldsymbol{x}|} \tag{3}$$

G Additional Probing Results

For a more detailed regression evaluation, we consider the Pearson correlation ρ, the coefficient of determination R^2 (which $= \rho^2$ for linear regression), mean square error (MSE), and approximate accuracy (AAcc).[5] Fig. 7 and Fig. 9 are expanded versions of Fig. 2 and Fig. 5.

The individual models' results for Fig. 3 can be found in Fig. 10.

H Additional Error Analysis

Here, we present additional experimental results and analysis to complement our discussion in §3.3.

[5] See §F for more details about metrics.

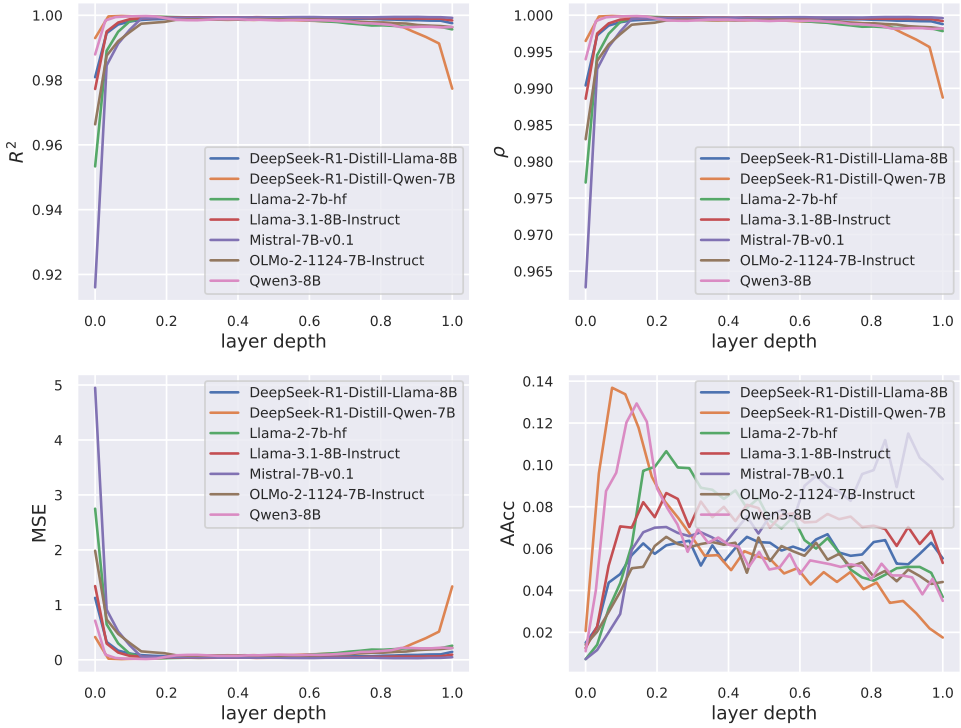

Figure 7: Regression probe performance on cross-notation data of each LLM across layers. Here, we report the Pearson correlation ρ, the coefficient of determination R^2 (which $= \rho^2$ for linear regression), mean square error (MSE), and approximate accuracy (AAcc). Definitions of MSE and AAcc can be found in §F. This is the expanded version of Fig. 2.

One-shot Prompting. We observed in §3.3 and Fig. 3 that Llama-2-7B and Mistral-7B ignore the values of the numbers and answer only with the second number. This may be because the example that we provided in the one-shot prompt (see §2 and §C) placed the correct answer in the second position. Thus, we tried exchanging the two numbers in the example. Fig. 11 displays the results of the altered prompt next to the original.

We observe that, out of all models, Llama-2's response to the change in prompt is the most extreme. In both prompts, Llama-2 answered exclusively with the same position in which the answer was presented in the one-shot example. Other models (DeepSeek-R1-Distill-Llama-8B, Mistral) show similar tendencies, but to a lesser extent. OLMo-2 shows a slight though consistent preference for the second number, regardless of the one-shot prompt. In contrast, GPT-4.1 and GPT-4.1-mini show minimal sensitivity to the ordering of the two options in the prompt, maintaining consistent performance across both variations.

Few-shot Prompting. Throughout this work, we have focused on one-shot prompting. We take the view that numeracy should be an innate capability: models should already exercise basic numeracy skills—cross-notation comparison being just one example—whenever they read scientific documents. In lieu of instructions, a single demonstration should be enough to illustrate the intended task and output format. Nevertheless, we evaluate few-shot performance here for completeness.

Fig. 12 shows the few-shot performance on cross-notation comparison using up to 5 examples (see specific few-shot prompts in §C). Overall, all models benefit from additional demonstrations in the few-shot prompt, with performance improvement plateauing around 5 examples. Still, all models struggle when the two numbers are close ($|\log_2(a/b)| < 0.1$). While the smaller models (7B–8B) perform around random chance in this regime, GPT-4.1 and GPT-4.1-mini maintain 75–85% accuracy, with additional demonstrations providing marginal further improvement.

We designed our 5 examples so that the correct answer's position alternates between the first place and the second place, in an attempt to eliminate positional bias we observed in Fig. 11. While Llama-2, OLMo-2 and Mistral's positional biases are remedied by additional examples, the positional bias of

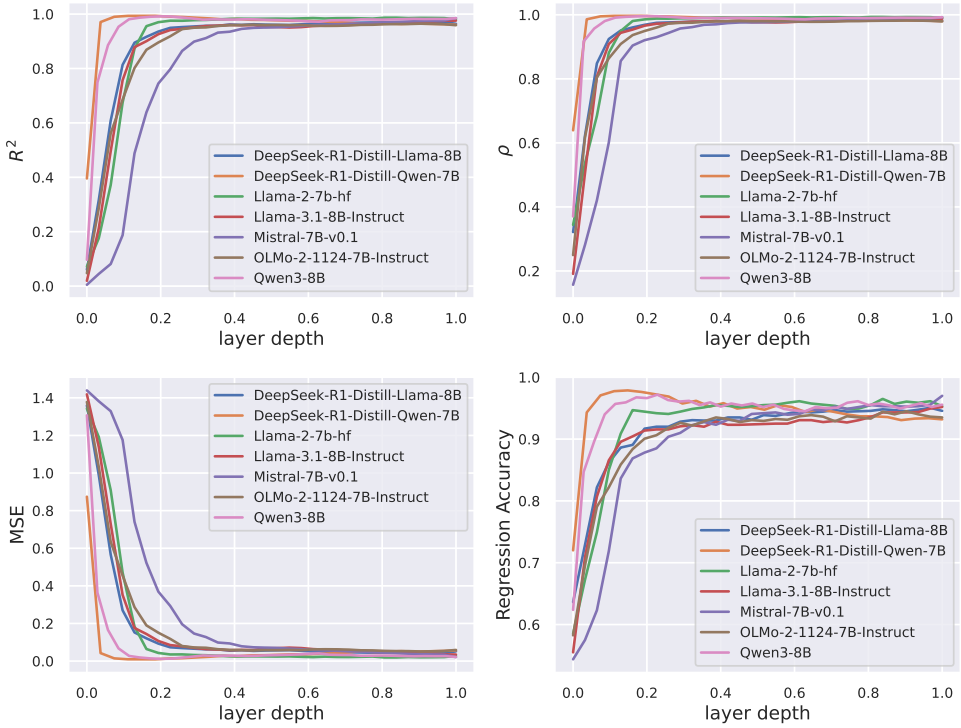

Figure 8: *Log-ratio* regression probe performance on cross-notation data of each LLM across layers. Here, we report the Pearson correlation ρ, the coefficient of determination R^2 (which $= \rho^2$ for linear regression), mean square error (MSE), and approximate accuracy (AAcc). Definitions of MSE and AAcc can be found in §F.

Deepseek-R1-Distill-Llama-8B remains.

Number of digits and magnitudes of numbers. Finally, we examine the effect of the number of digits and magnitudes of numbers on cross-notation comparison via either verbalization or probes. Fig. 13 and Fig. 14 plot accuracy with respect to number of digits and magnitude (measured by $\log_2(a+b)$), respectively. We observe that verbalized comparison accuracy generally deteriorates for numbers that are larger or have more digits in their surface form. In contrast, comparison accuracy via regression or logistic classifier remains much higher and does not deteriorate in either case. These observations suggest that LLMs implicitly understand larger and longer numerals but fail to verbalize their relationship.

I Computational Budget

All our experiments used a single NVIDIA A100 GPU (40GB or 80GB). Extracting hidden states for the full dataset takes a few hours, training and evaluating probes for one model takes about 40 minutes, and verbalization evaluation on 1,600 examples takes roughly 25 minutes per model. Finetuning a single model under one hyperparameter setting takes around 1 hour.